



Elective - III

UE23AM342BA3 : Explainable AI

Banana Problem – Unit 2

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Banana Problem –Unit2 Individual Activity (10 Marks)

Problem Title 1 (For odd SRNs):

Global Feature Effect Exploration using built-in dataset.

Dataset

Use **Breast Cancer Dataset** from **sklearn**

Objective :

Experimentally compare PDP vs ALE vs ICE and study feature interaction + correlation effects.

1. Tasks to be Performed:

1. Load breast cancer dataset from sklearn
2. Train Black Box Model(Random Forest / XGBoost / Neural Network).
3. Select Features for Explanation (select 2 highly important features (from model importance or SHAP) 1 moderately important feature)
4. Generate Explanations(must implement PDP, ICE, ALE(use PyALE library))
5. Perform following Experiments:
 - a) Test Feature Correlation Effect(Pick 2 correlated features, then compare PDP vs ALE)
 - b) Subgroup Behaviour(Check ICE curves → Are all samples behaving same?)
 - c) Model Change Effect(Train second model → Compare PDP shapes)

Banana Problem –Unit2 Individual Activity (10 Marks)



Problem Title 2 (For even SRNs):

Feature Contribution + Recourse Analysis using built-in dataset.

Dataset

Use **California Housing Dataset** from **sklearn**

Objective :

Perform feature contribution and recourse analysis using California Housing Dataset.

Tasks to be Performed:

1. Load California Housing dataset from sklearn
2. Train the model (Random Forest or XGBoost regressor. Find RMSE or R^2).
3. Perform SHAP Analysis (must generate global importance(`shap.plots.bar()`), Summary Plot(`shap.summary_plot()`) and Local Explanation(`shap.plots.waterfall()`))
4. Generate Counterfactual Explanations :
 Use DiCE to generate counterfactuals such that:
 Increase predicted house price by X amount
 (OR) *Reduce predicted house price*
4. Experimental What-If Studies: Must perform
 - a) Actionable Counterfactual(Allow change only in *income* and *house age*)
 - b) Realistic Constraint Counterfactual (Fix *latitude* and *longitude*)

Students must submit:

- ✓ One case where SHAP importance $\neq$ counterfactual change
- ✓ One surprising feature behaviour
- ✓ One model bias observation

Submission Link

<https://forms.gle/bnUF9zrtuuP77tGT6>



THANK YOU
